Prey-Predator Maze with Reinforcement Learning

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***Abstract*—The paper explores the study and design of a complex system consisting of a prey and predator inside a maze, each has owner objective, survive and hunt respectively. Using systems thinking for create a model of this system. Designed a proposal with two cybernetic agents that will use reinforcement learning to maximize the prey and predator objective in a competitive scenery. This project search explores and investigate if this model is effective with the use of AI and cybernetic foundations.**

***Index Terms*—Reinforcement Learning, Multi-Agent Systems, Prey-Predator, Cybernetics, Maze Simulation**

I. Introduction

Prey and predator is a system that is often seen in nature. It consists in the interaction between species (predator) and other species (prey), where the first searches to hunt the second. On a large scale, it is essential for an ecosystem because the predator populations regulate the prey populations, like in the Lotka-Volterra model [1]. On a small scale, it allows studying the behavior of each specimen.

The system that this project focuses on includes one specimen of each species. The prey's goal is to survive as long as possible. The predator's goal is to hunt the prey as quickly as possible. The interaction between both is competitive because they pursue opposite objectives.

This kind of system is present in various ecosystems like the sea, the jungle, or closed spaces for experiments. The second important part of this system is the environment—a maze. A maze is a place artificially formed by paths and crossroads designed to confuse the player inside. Normally, in a maze, the player searches for an objective, like the exit or a specific item. In this environment, each individual has a different goal; they act like players inside the maze. The predator walks the maze in search of the prey. The prey, however, has no tangible objective—its goal is to survive and avoid the predator.

The labyrinth has two inherent components: walls and paths. The paths are perfectly passable, while the walls cannot be passed through. In a maze, it is not possible to see far because visibility defeats its function. A maze can also include other elements like traps—zones where the specimen can fall, losing the possibility of completing their objectives.

The previously described system is complex and poses a non-trivial problem when trying to emulate it computationally. Some examples are described in [2], where a solution is presented using Multi-Agent Reinforcement Learning and Game Theory for decision-making. Another case is shown in [3], where Reinforcement Learning is applied in Two-Player Zero-Sum scenarios, as both agents pursue opposite objectives and receive rewards equivalent to the loss of the other.

To define a clear scenario, the prey will be taken as a rabbit and the predator as a fox. Both are recognizable specimens and fulfill the dynamic, as the fox is a natural predator of the rabbit. The real-world characteristics of these animals can affect their interaction. Some rules will be taken into account: the rabbit is fast, but not faster than the fox. As seen in [4], a main advantage of the rabbit is its maneuverability compared to the fox. In [5], the rabbit's ability to avoid traps is explored—an important trait. On the fox's side, it is faster than the rabbit and can increase its speed when approaching the objective, as studied in [6]. The fox also has a keen sense of smell [7], greater than the rabbit's, which helps it infer where the prey is.

II. Method and Materials

Creating the model required a complete analysis and design of the system, using systems thinking to identify the parts and their interrelations.

There are two main actors in this system: the prey and the predator. They both have a competitive relationship, and the behavior of one directly influences the behavior of the other. If the predator sees or smells the prey, it will switch to hunting mode. This mode represents an increase in speed and triggers a directional instinct for catching the prey—an internal behavior similar to natural instinct. The predator's speed can be helpful or harmful depending on the environment; since it includes direction changes, the added speed can help catch the prey quickly or obstruct its movement.

On the other hand, the prey can see the predator and remain cautious. The prey seeks to escape by moving in the opposite direction or making erratic movements to confuse the predator. Additionally, the prey leaves behind a residual smell that fades over time. Although the prey does not have as many capabilities as the predator, its maneuverability gives it an advantage in the maze scenario.

Other elements also affect the main actors' interactions. Walls block movement and vision. Traps are elements that can capture the prey, preventing either agent from achieving their goals. Traps are chaotic components in the system since they appear in random locations. A trap near the prey at the beginning can end the simulation quickly and negatively impact the expected result. Paths are the only safe zones where agents can move freely.

With the elements defined, a high-level structure of the system was created, based on cybernetic principles. The focus is on control and communication. Each agent has sensors or abilities that collect information and generate inputs to the system. This information is processed and evaluated through rewards to make decisions. Based on the decision, the agent's actuators perform actions that affect the state of the environment.

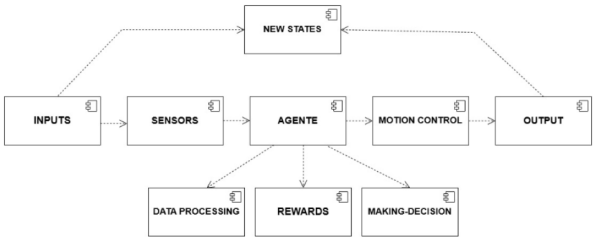

Fig. 1. Estructura de alto nivel del sistema.

As shown in Figure 1, each agent can receive and process information in different ways. Feedback loops are essential in the model, as they form part of the control mechanism. When an agent receives information and performs an action, it receives a reward that teaches it the expected behavior. A negative reward signals that something was wrong, and the agent will likely choose a different action in the future. A positive reward reinforces good behavior.

TABLE I
TABLE 1. REWARDS AND PENALTIES OF THE PREY.

Event	Reward
Captured by predator	-0.5
Fell in a trap	-0.5
Each step alive	+0.1
Not moving	-0.05

TABLE II
TABLE 2. REWARDS AND PENALTIES OF THE PREDATOR.

Event	Reward
Captures the prey	-0.5
Approaches the prey when seen	-0.5
Moves away from the prey when seen	+0.1
Explores by moving	+0.05
Not moving	-0.05

This model represents the maze as a 10x10 grid, with agents as elements occupying cells. There are three different matrices representing the environment's state. The state matrix represents the scene, with walls, paths, and traps. It changes at the start of each simulation since traps are randomly placed. The agent matrix tracks agent positions and changes dynamically with movement. The smell matrix holds the residual scent left by the prey, which fades over time.

Both agents have the same actuators for motion control: four discrete actions—up, down, left, and right. Movements update the matrices, and reactions differ depending on what is encountered. Agents may move into a wall (blocked), over a trap (danger), or on a path (safe). They cannot move outside the maze. They may also move to a position occupied by the other agent. Additionally, the prey's movement affects the smell matrix: each time it moves, it leaves a scent value of 5 at its location, which gradually decays to 0 over time.

The model includes a proximity sensor, representing the prey's vision. The sensor evaluates surroundings in horizontal, vertical, and diagonal directions, with a 5-cell range in each direction. The prey uses this to construct an observation vector with coordinates and numerical values for each element, as described in Table III.

TABLE III
TABLE 3. ENVIRONMENT ELEMENTS.

Code	Element
1	Prey
2	Predator
3	Wall
4	Trap
5	Path
0	No vision

The predator uses the same sensor when in normal mode. In hunting mode, however, the predator switches to a faster sensor focused on locating the prey. The predator also has a sense of smell. It uses this to detect the intensity of scent and identify where the strongest signal comes from using the smell matrix. This data is compiled into a vector similar to the prey's, but with additional elements such as whether hunting mode is active and the maximum detected smell with its coordinates.

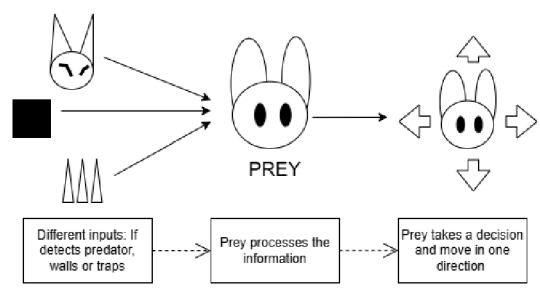

Fig. 2. Phase portrait of the prey.

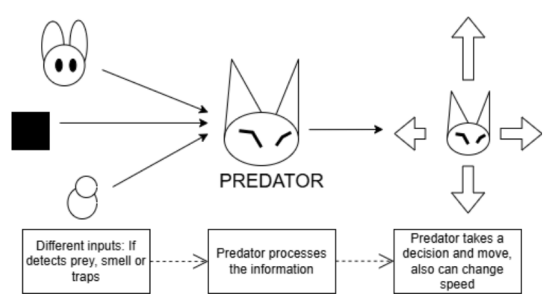

Fig. 3. Phase portrait of the predator.

This system includes a dynamic model, with various elements changing the environment state over time, starting with the agents' movements. Each agent constantly changes location as it seeks to fulfill its objective. Additional factors also evolve, such as the probability of the prey avoiding traps (which increases with time) or the predator's decision to enter hunting mode and speed up. These variables introduce chaos into the system and make the agents' behavior hard to predict.

TABLE IV
TABLE 4. EQUATIONS OF THE DYNAMIC MODEL.

Equation	Detailed Explanation
$\pi_p(t+1) = \pi_p(t) + \Delta t\, r_p(t)\, v_p(t)$	Updates predator position by adding the product of time-step Δt, predator speed $v_p(t)$, and movement direction $r_p(t)$ (a unit vector) to the current position $\pi_p(t)$.
$\pi_h(t+1) = \pi_h(t) + \Delta t\, r_h(t)\, u_h(t)$	Updates prey position using analogous terms: current position $\pi_h(t)$, time-step Δt, movement direction $r_h(t)$, and speed $u_h(t)$.
$v_p(t) = \nu_{\max} + \frac{\alpha}{1+e^{-kd(t)}}$	Models predator speed adaptation: base speed $\nu_{\max}$ plus a logistic function scaled by α, where k controls sensitivity to prey distance $d(t)$, resulting in acceleration when the prey is near.
$P_{\text{evade}}(t) = P_{\text{base}}\left(1 - e^{-\beta t}\right)$	Defines probability of successful evasion for the prey: starts at zero and asymptotically approaches P_{base}, with learning rate governed by β, simulating experience-based evasion improvement.

This model is agent-based, as the behavior is discrete rather than continuous. There are no suitable differential equations to represent the actions of the prey and predator. Therefore, an agent-based approach is more appropriate to observe how learning emerges over time. Implementing sensors, actuators, and observations has proven effective for simulating this system and evaluating agent performance in the environment.

Finally, the designed model is implemented with two machine learning models.

One of these models determines whether to activate hunting mode. The predator uses this decision model, trained with 20,000 synthetic data samples, based on two features: 'Seen' (a Boolean indicating whether the prey is visible) and 'Intensity' (the strongest scent detected). Data was generated with probabilistic variation to reflect realistic behavior—sometimes the predator does not attack head-on or may be fatigued and chooses not to run.

The dataset was split into 80% for training and 20% for testing. Two supervised learning algorithms were compared: Logistic Regression (useful for binary classification) and Gradient Boosted Decision Trees (GBDT). The best-performing model was used in the predator's implementation. The chosen model is responsible for determining when to activate hunting mode based on sensor and scent data.

To build the maze environment, PettingZoo Parallel with Gymnasium was used to support competitive multi-agent environments with parallel agent actions. Several Python libraries were used to implement the environment; Pygame was used for visual rendering.

For agent training, the PPO (Proximal Policy Optimization) algorithm was applied. This algorithm is stable and suitable for multi-agent learning, using two policies per agent and observations as inputs to a neural network. After a predefined number of episodes, training is completed and saved as a checkpoint. The trained model is then used for result visualization.

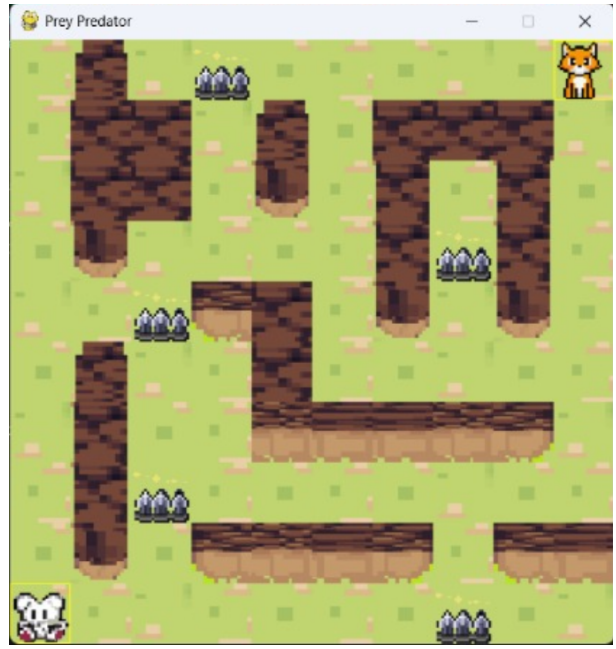

Fig. 4. Environment visualization using Python tiles.

III. RESULTS

Different experiments were carried out to evaluate the model's performance. Initially, supervised machine learning was used to predict the predator's hunting mode. Both models (Logistic Regression and Gradient Boosted Decision Trees) were tested on three datasets using various metrics. While accuracy measures the number of correct responses, it is not reliable in this case due to potential data imbalance. Instead, the F1 score and AUC-ROC provided better insight for deciding when to activate hunting mode.

The results showed that the Gradient Boosted Decision Tree consistently performed better across all metrics and datasets, proving to be more reliable. Therefore, it was chosen to drive the predator's hunting behavior using sensor and scent information.

TABLE V
TABLE 5. RESULTS OF LOGISTIC REGRESSION MODEL.

Dataset	Accuracy	F1 Score	AUC-ROC
1	0.8619	0.7903	0.5310
2	0.8601	0.7892	0.5255
3	0.8643	0.7846	0.5362

TABLE VI
TABLE 6. RESULTS OF GRADIENT BOOSTED DECISION TREE MODEL.

Dataset	Accuracy	F1 Score	AUC-ROC
1	0.8700	0.9043	0.9080
2	0.8753	0.9076	0.9160
3	0.8765	0.9085	0.9162

In the reinforcement learning experiments, different training settings were tested by modifying the reward structure and

algorithm. Initially, complex and abundant rewards were used. These worked well during early training, as the predator learned to hunt efficiently. However, after many episodes, the predator learned to exploit rewards by staying close to the prey rather than capturing it.

The prey's behavior also varied. When traps were heavily penalized, the prey preferred not to move. When penalization for not moving was too high, the prey moved freely, even if it risked stepping into traps.

After testing simpler reward configurations, prey behavior improved slightly. It learned to move and gain rewards, but the predator remained mostly passive—roaming its area without actively seeking or catching the prey, indicating the reward system still lacked proper incentives.

The final reward setup (as defined earlier) produced more convergent behavior, but still not the expected strategic interaction. The prey began showing reward-seeking behavior by exploring movement, while the predator maintained a baseline reward through idle or exploratory actions.

The following figures show maximum rewards obtained by the agents after increasing training episodes. During the first 100 iterations, both agents mostly explored randomly.

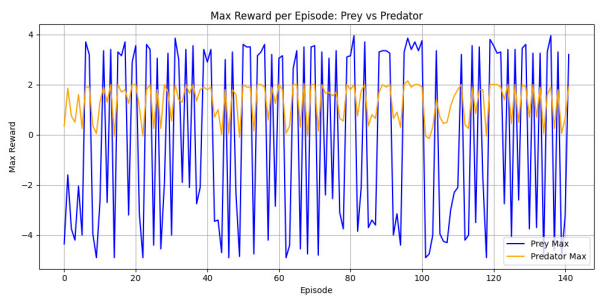

Fig. 5. Figure 5. Evaluation with 100 episodes.

After 1000 episodes, the prey began to stay still—resulting in consistent failure. Meanwhile, the predator continued to explore a single zone, possibly gaining minor rewards from exploration that offset the lack of hunting.

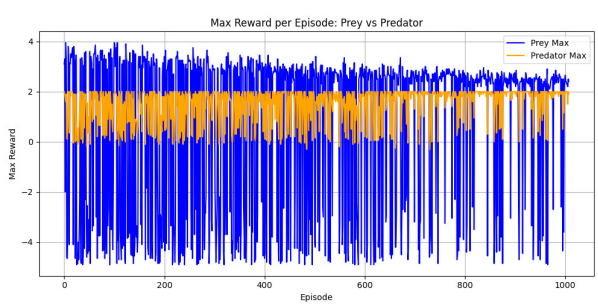

Fig. 6. Figure 6. Evaluation with 1000 episodes.

At 2000 episodes, the same behavior persisted. The prey's inactivity reduced its reward. The predator maintained a positive reward with minimal movement, likely due to prior punishment for failing to catch the prey.

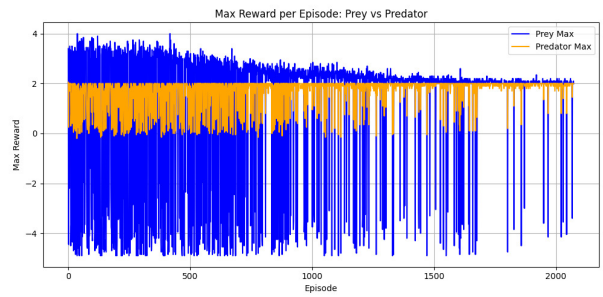

Fig. 7. Figure 7. Evaluation with 2000 episodes.

By 4000 episodes, a significant change occurred. The prey started moving in multiple directions, avoiding punishment for idleness and receiving larger rewards.

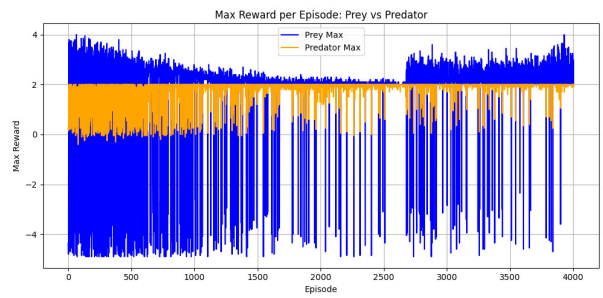

Fig. 8. Figure 8. Evaluation with 4000 episodes.

In the final training with 8000 episodes, the prey continued to move and maintained high reward levels. The predator's behavior remained consistent with earlier patterns.

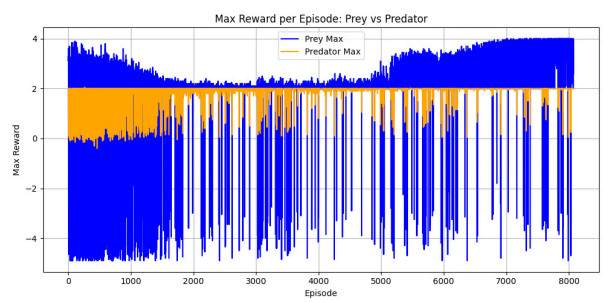

Fig. 9. Figure 9. Evaluation with 8000 episodes.

IV. Conclusions

The proposed model serves as a valid approach to explore the behavior of two cybernetic agents within a prey-predator system. Although it exhibited convergence in behavior, the outcome did not fully align with the expected dynamics. The prey learned to maximize its reward by staying alive and avoiding penalties for inaction. During training, it explored various alternatives and eventually adopted movement as a strategy to maintain high rewards.

The predator, meanwhile, maintained a consistent reward, gradually learning how to gather points without necessarily hunting the prey. This behavior indicated a gap in the observational space and reward structure.

One of the main failures of the project was the observation vector. Different reward structures were implemented for each agent, but the predator never actively sought to capture the prey. The observation vector was too large and poorly structured, preventing the predator from understanding where to find the prey or why catching it would yield a high reward.

Despite these limitations, the project succeeded in offering a platform to explore and study prey-predator interactions. Future improvements should focus on refining the observation space, reward shaping, and encouraging strategic agent behavior through better sensor design and dynamic adaptation.

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